

A Comparative Catalogue of Gravitational and Orbital Time-Dilation Effects in Galactic X-ray Binary Systems

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Abstract

General relativity predicts that proper time runs at different rates depending on an observer’s position within a gravitational potential and on their state of motion. While these effects are minuscule in terrestrial and Solar System contexts, they become significant in the vicinity of stellar-mass compact objects such as black holes and neutron stars. In this work we compile and analyse a homogeneous sample of 18 well-characterised Galactic X-ray binaries — comprising 9 black-hole and 9 neutron-star systems, predominantly low-mass X-ray binaries (LMXBs) together with a small number of intermediate- and high-mass systems — drawing on published dynamical mass measurements, orbital periods, and distances. For each system we compute (i) the Schwarzschild radius of the compact object, (ii) the gravitational time-dilation factor for a notional probe held at a fixed altitude of 1000 km above this radius, and (iii) the combined special- and general-relativistic time-dilation experienced by the donor star due to its orbital velocity and its position in the compact object’s gravitational potential. For a subset of five black holes with published spin measurements, we additionally compute (iv) the spin-dependent proper-time deficit for matter in a circular orbit at the Kerr innermost stable circular orbit (ISCO), and (v) perform a Monte Carlo propagation of mass uncertainties to obtain credible intervals on all derived quantities. We find that donor stars in the tightest, most compact systems (e.g. 4U 1916-053, XTE J1118+480, A0620-00) experience total proper-time lags of order 1–10 parts per million relative to a distant static observer, corresponding to roughly 5–8 hours per century, with Monte Carlo relative uncertainties typically of 5–10%. For a hypothetical probe parked 1000 km above the Schwarzschild radius of the most massive black hole in our sample, Cygnus X-1, the gravitational time lag would amount to approximately 3 years per century relative to Earth. For the same object, whose spin is measured to be near-extremal ($a_* \approx 0.998$), the proper-time deficit for matter orbiting at the Kerr ISCO reaches $\approx 32\%$, compared to $\approx 5.7\%$ for a non-spinning black hole at the equivalent orbital radius — demonstrating that spin is the dominant factor governing time dilation in the strong-field regime immediately outside the horizon. These results provide a quantitative, observationally grounded benchmark for the scale of relativistic time-dilation effects accessible in principle within the Milky Way, and may serve as reference values for outreach material and for theoretical studies of relativistic clock networks around compact objects.

1 Introduction

The dilation of time in strong gravitational fields and at relativistic velocities is among the most striking and best-tested predictions of Einstein’s general theory of relativity (Misner et al., 1973). Near the Earth and within the Solar System, these effects are tiny — of order parts in 10^{10} — yet they are routinely measured and corrected for, most famously in the operation of the Global Positioning System. Far larger effects occur in the vicinity of stellar-mass compact objects: black holes (BHs) and neutron stars (NSs) that are the end-points of massive-star evolution and that frequently reside in close binary systems with ordinary stars, forming X-ray binaries (Shapiro and Teukolsky, 1983; Liu et al., 2007).

Galactic X-ray binaries, particularly low-mass X-

ray binaries (LMXBs) in which a compact object accretes material from a low-mass donor star via Roche-lobe overflow, are among the best-studied laboratories for compact object astrophysics. Decades of optical, infrared, and X-ray monitoring have yielded precise dynamical masses, orbital periods, and distances for dozens of systems (Casares and Jonker, 2014; Corral-Santana et al., 2016; Fortin et al., 2024). These measured parameters make it possible to calculate, for each system, the magnitude of gravitational and orbital time-dilation effects using only well-established general-relativistic formulae — without invoking any speculative physics.

The motivation for this study is twofold. First, from a pedagogical and outreach perspective, real astrophys-

ical systems provide far more compelling and concrete benchmarks for relativistic time-dilation than purely hypothetical scenarios: a popular fascination with the idea of “time travel” can be channelled into an accurate appreciation of where, within our own Galaxy, the largest *naturally occurring* time-dilation effects are actually found. Second, from a research perspective, a homogeneous catalogue of derived time-dilation quantities — alongside the dynamical parameters from which they are computed — provides a convenient reference for studies of relativistic clock networks, forecasts for future timing missions near compact objects, and comparative analyses of how time-dilation scales with compact-object mass and orbital compactness across a population of systems.

In this paper we compile a sample of 18 Galactic X-ray binaries with well-constrained compact-object masses, donor masses, orbital periods, and distances (Section 2). We then compute, for each system, the Schwarzschild radius of the compact object, the gravitational time-dilation factor at a fixed reference altitude above this radius, and the combined gravitational-plus-orbital time-dilation of the donor star (Section 3). We present the resulting catalogue and discuss trends with compact-object mass and binary compactness (Section 4), before concluding with a discussion of caveats and possible extensions (Section 5).

2 Data

We assembled a sample of 18 Galactic X-ray binary systems with dynamically measured compact-object masses, drawing on the compilations of Casares and Jonker (2014), Liu et al. (2007), Corral-Santana et al. (2016), Fortin et al. (2024), and Avakyan et al. (2023), supplemented by individual mass and distance determinations for specific systems (Miller-Jones et al., 2021, 2009; Reid et al., 2014; Steeghs et al., 2013; Galloway et al., 2008). The sample comprises 9 black-hole and 9 neutron-star systems: Cygnus X-1, V404 Cygni, A0620-00, GRO J1655-40, XTE J1118+480, GX 339-4, 4U 1543-475, XTE J1550-564, and GRS 1915+105 (black holes); and Cygnus X-2, Sco X-1, Cen X-4, 4U 1822-371, XTE J2123-058, GS 1826-238, 4U 1916-053, EXO 0748-676, and 4U 1636-536 (neutron stars). The majority of these are low-mass X-ray binaries (LMXBs), in which the donor is a low-mass ($\lesssim 1 M_\odot$) star transferring mass via Roche-lobe overflow. Two notable exceptions are intermediate-mass systems: Cygnus X-2, whose donor is an evolved subgiant with an estimated current mass of $\sim 0.5\text{--}0.7 M_\odot$ descended from a more massive ($\sim 3.5 M_\odot$) progenitor (Casares et al., 2010), and GRO J1655-40, whose F/G-type subgiant donor has a mass of $\sim 2.6 M_\odot$ (Beer and Podsiadlowski, 2002), placing it in the intermediate-mass X-ray binary

(IMXB) regime. Cygnus X-1 is a qualitatively different case: its donor, HDE 226868, is an O-type supergiant of mass $\sim 40.6 M_\odot$ (Miller-Jones et al., 2021), making Cygnus X-1 a high-mass X-ray binary (HMXB) rather than an LMXB. We retain Cygnus X-1 in our sample because the time-dilation formalism of Section 3 depends only on the dynamical masses, orbital period, and distance — not on the evolutionary classification of the donor — and because Cygnus X-1 hosts the most massive, and hence most strongly time-dilating, compact object in our sample. We use the term “X-ray binaries” throughout to refer to the sample as a whole, and refer to individual LMXB, IMXB, or HMXB classifications only where relevant.

For each system we record: the compact-object mass M_X (in solar masses, $M_\odot$), the donor-star mass M_d , the orbital period P_{orb} , the distance d , and the orbital inclination i . For systems with multiple published mass determinations, we adopted the value preferred by the most recent dynamical study cited above. Where the compact object is a neutron star without an individually measured mass, we adopted the canonical value $M_X = 1.4 M_\odot$, following standard practice (Galloway et al., 2008). The full input table is provided in `data/dataset.csv`.

We emphasise that this sample is not intended to be exhaustive or statistically complete; rather, it represents a selection of well-studied, frequently cited systems spanning roughly an order of magnitude in compact-object mass ($1.4\text{--}21.2 M_\odot$) and three orders of magnitude in orbital period (0.03–33.5 days), which is sufficient to illustrate the scaling of relativistic time-dilation effects with these parameters.

3 Methods

3.1 Schwarzschild radius

For each compact object of mass M_X , we compute the Schwarzschild radius

$$R_s = \frac{2GM_X}{c^2}, \quad (1)$$

where G is the gravitational constant and c is the speed of light in vacuum. This is the characteristic length scale of the compact object’s gravitational field and sets the scale for all subsequent time-dilation calculations.

3.2 Gravitational time dilation at a fixed altitude

For a static observer at Schwarzschild radial coordinate $r > R_s$, the rate of proper time τ relative to the coordinate time t of a distant static observer is

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{R_s}{r}}. \quad (2)$$

To compare compact objects of very different masses on a common, physically meaningful footing, we evaluate Eq. (2) at a fixed *altitude* of 1000 km above the Schwarzschild radius, i.e. $r = R_s + 10^6$ m. This choice represents a notional probe hovering just outside the compact object at a distance that is observationally and operationally meaningful (comparable to low orbital altitudes around Earth), while remaining safely outside R_s for all objects in our sample. We report the resulting fractional time lag,

$$\Delta = 1 - \frac{d\tau}{dt}, \quad (3)$$

in parts per million (ppm) and in equivalent days per century.

We note that the time-dilation factor evaluated *at the innermost stable circular orbit* (ISCO), $r_{\text{ISCO}} = 3R_s$ for a non-rotating (Schwarzschild) black hole, takes the universal, mass-independent value $\sqrt{2/3} \approx 0.8165$ (Bardeen et al., 1972). This well-known result means that, in terms of *fractional* time dilation, the ISCO of *any* non-spinning black hole — regardless of its mass — imposes the same $\sim 18.4\%$ proper-time deficit relative to a distant observer. For this reason we use the fixed-altitude prescription above, rather than the ISCO, to obtain a mass-dependent diagnostic.

3.3 Donor-star orbital and gravitational time dilation

For each binary, we compute the relative semi-major axis a_{rel} of the two-body orbit from Kepler’s third law,

$$a_{\text{rel}} = \left(\frac{GM_{\text{tot}} P_{\text{orb}}^2}{4\pi^2} \right)^{1/3}, \quad (4)$$

where $M_{\text{tot}} = M_X + M_d$ is the total system mass and P_{orb} is the orbital period. The donor star’s orbital radius about the system barycenter is then $a_d = a_{\text{rel}} M_X / M_{\text{tot}}$, and its circular-orbit speed is

$$v_d = \frac{2\pi a_d}{P_{\text{orb}}}. \quad (5)$$

The donor star experiences two distinct relativistic effects relative to a distant static observer: (i) special-relativistic time dilation due to its orbital velocity,

$$\left(\frac{d\tau}{dt} \right)_{\text{SR}} = \sqrt{1 - \frac{v_d^2}{c^2}}, \quad (6)$$

and (ii) general-relativistic time dilation due to its position in the compact object’s gravitational potential. Since $a_d \gg R_s$ for all systems in our sample (the donor orbits at distances of order 10^8 – 10^{11} m, compared to $R_s \sim 10^4$ – 10^5 m), we use the weak-field approximation

$$\left(\frac{d\tau}{dt} \right)_{\text{GR}} \approx 1 - \frac{GM_X}{a_d c^2}. \quad (7)$$

To leading order, the two fractional lags are additive, and we define the total fractional time lag of the donor star as

$$\Delta_{\text{tot}} = \left(1 - \left(\frac{d\tau}{dt} \right)_{\text{SR}} \right) + \left(1 - \left(\frac{d\tau}{dt} \right)_{\text{GR}} \right). \quad (8)$$

We report Δ_{tot} in parts per million and as an equivalent lag in hours per century.

3.4 Kerr (spinning) black holes: time dilation at the ISCO

The calculations of Sections 3.1–3.3 assume a non-rotating (Schwarzschild) compact object. However, all astrophysical black holes are expected to possess non-zero angular momentum, parameterised by the dimensionless spin $a_* = a/M \in [-1, 1]$ (with $a_* > 0$ denoting prograde rotation relative to the orbit under consideration). For a subset of black holes in our sample, a_* has been measured via the continuum-fitting method, which fits the thermal X-ray spectrum of the accretion disk to a relativistic thin-disk model and identifies the inner disk edge with the ISCO (Shafee et al., 2006; McClintock et al., 2006; Gou et al., 2010, 2011; Zhao et al., 2021).

For a Kerr black hole, the ISCO radius of a prograde equatorial circular orbit is given by the Bardeen–Press–Teukolsky formula (Bardeen et al., 1972):

$$r_{\text{ISCO}} = M \left[3 + Z_2 - \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)} \right], \quad (9)$$

where M is expressed in gravitational units ($r_g \equiv GM/c^2$) and

$$Z_1 = 1 + (1 - a_*^2)^{1/3} \left[(1 + a_*)^{1/3} + (1 - a_*)^{1/3} \right], \quad (10)$$

$$Z_2 = \sqrt{3a_*^2 + Z_1^2}. \quad (11)$$

For $a_* = 0$, Eq. (9) reduces to the familiar Schwarzschild result $r_{\text{ISCO}} = 6r_g = 3R_s$; for a maximally spinning prograde black hole ($a_* \rightarrow 1$), $r_{\text{ISCO}} \rightarrow r_g$.

The time-dilation factor experienced by matter in a circular orbit at r_{ISCO} is given by the specific orbital energy $\tilde{E} = -u_t$, which for a circular equatorial Kerr orbit is (Bardeen et al., 1972)

$$\tilde{E}(r) = \frac{r^{3/2} - 2r^{1/2} + a_*}{r^{3/4} \sqrt{r^{3/2} - 3r^{1/2} + 2a_*}}, \quad (12)$$

with r expressed in units of r_g . This quantity equals $d\tau/dt$ for the orbiting matter and combines both the gravitational and orbital-velocity contributions to the time dilation, analogous to Δ_{tot} for the donor star but

evaluated at the ISCO itself. We report the fractional proper-time deficit at the ISCO, $1 - \tilde{E}(r_{\text{ISCO}})$, as a function of a_* .

For reference, evaluating Eq. (12) at the Schwarzschild ISCO ($a_* = 0$, $r = 6r_g$) gives $\tilde{E} = \sqrt{8/9} \approx 0.9428$, i.e. a $\sim 5.7\%$ proper-time deficit. This is distinct from, and should not be confused with, the $\sim 18.4\%$ *static-observer* redshift $\sqrt{1 - R_s/r}$ at $r = 3R_s$ discussed in Section 3.2: the latter applies to a hypothetical observer held *stationary* at the ISCO radius (which is not itself a physically realisable circular orbit), whereas Eq. (12) applies to matter *actually orbiting* at that radius.

3.5 Monte Carlo uncertainty propagation

To address the limitation, discussed further in Section 5, that observational uncertainties in M_X and M_d propagate into Δ_{1000} and Δ_{tot} , we perform a Monte Carlo uncertainty propagation for all 18 systems in the sample. For each system, we draw $N = 10^5$ samples of M_X and M_d from independent normal distributions centred on the adopted values (Table S1 of the Supplementary Material, also available as `data/dataset.csv`) with standard deviations given by the published 1σ uncertainties where available. For the nine neutron-star systems with the canonical $M_X = 1.4 M_\odot$ (for which no individual dynamical uncertainty is available), we adopt $\sigma_{M_X} = 0.2 M_\odot$, consistent with the scatter in dynamically measured neutron-star masses in LMXBs (Galloway et al., 2008). For donor masses without a published uncertainty, we adopt a fractional uncertainty of 15%. The orbital period P_{orb} is treated as exact, since its fractional uncertainty is negligible ($\lesssim 10^{-4}$) for all systems in our sample. For each Monte Carlo sample we recompute Δ_{1000} and Δ_{tot} via Eqs. (3) and (8), and report the median and the 16th–84th percentile range (approximately a 1σ credible interval for a Gaussian) for each system.

All calculations were performed in Python using `numpy` and `pandas`; the complete analysis scripts are provided in `code/analysis.py` (Sections 3.1–3), `code/kerr_analysis.py` (Section 3.4), and `code/error_propagation.py` (Section 3.5). The full set of input and derived quantities is provided in `data/dataset.csv`, `data/derived_quantities.csv`, `data/spin_subset.csv`, `data/kerr_isco_results.csv`, and `data/monte_carlo_results.csv`.

4 Results

4.1 Schwarzschild radii and fixed-altitude time dilation

Table 1 summarises the key derived quantities for all 18 systems. The Schwarzschild radii range from $R_s \approx 4.13$ km for the canonical $1.4 M_\odot$ neutron stars to $R_s \approx 62.6$ km for the $21.2 M_\odot$ black hole in Cygnus X-1 (Miller-Jones et al., 2021). The corresponding fractional time-dilation lags for a probe held at a fixed altitude of 1000 km above R_s range from approximately 2.1×10^{-3} (2061 ppm) for the neutron-star systems to 3.0×10^{-2} (29 908 ppm, i.e. about 3%) for Cygnus X-1. Expressed as an accumulated lag, the latter corresponds to approximately 1092 days — roughly three years — per century relative to a distant observer (Fig. 2).

This result illustrates a simple but instructive point: among the systems considered, the single location within the Milky Way where a (hypothetical, non-destructible) clock would run measurably slower than on Earth, at an operationally meaningful altitude, is in the immediate vicinity of the most massive stellar black holes such as Cygnus X-1 — not near neutron stars, whose smaller Schwarzschild radii produce proportionally smaller time-dilation at the same fixed physical altitude.

4.2 Donor-star time dilation

Figure 1 shows the total donor-star time-dilation fraction Δ_{tot} (Eq. 8) as a function of compact-object mass M_X , separated by compact-object type. The values span roughly an order of magnitude, from $\Delta_{\text{tot}} \approx 0.24$ ppm for Cygnus X-2 (the widest-orbit neutron-star system in our sample, with $P_{\text{orb}} = 9.84$ days) to $\Delta_{\text{tot}} \approx 9.5$ ppm for XTE J1118+480, an ultracompact black-hole binary with $P_{\text{orb}} = 0.17$ days.

The dominant driver of Δ_{tot} is not compact-object mass per se, but orbital compactness: systems with the shortest orbital periods — and hence the smallest donor semi-major axes a_d — exhibit the largest values of both the special-relativistic (v_d/c) and general-relativistic (GM_X/a_dc^2) contributions. This is evident from the position of 4U 1916-053 ($P_{\text{orb}} = 0.0345$ days, $\Delta_{\text{tot}} \approx 8.9$ ppm) and XTE J1118+480 ($P_{\text{orb}} = 0.17$ days, $\Delta_{\text{tot}} \approx 9.5$ ppm) near the top of Fig. 1, despite both systems hosting comparatively low-to-moderate compact-object masses. Conversely, GRS 1915+105, which hosts one of the most massive black holes in the sample ($M_X = 12.4 M_\odot$) but has by far the widest orbit ($P_{\text{orb}} = 33.5$ days), exhibits the smallest total lag in the black-hole subsample, $\Delta_{\text{tot}} \approx 0.39$ ppm.

For the most extreme system in our sample, XTE J1118+480, the total donor-star lag of ~ 9.5 ppm corresponds to approximately 8.3 hours per century —

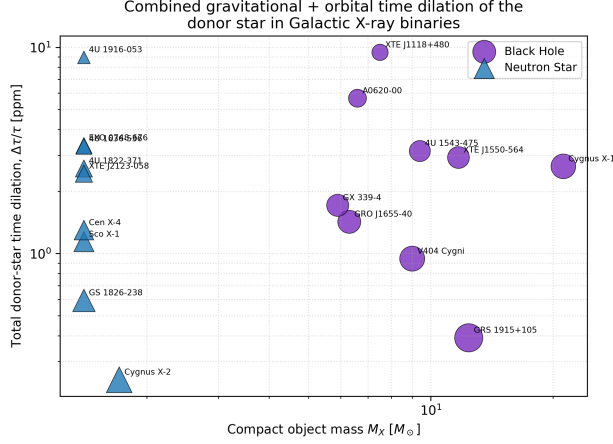


Figure 1: Total donor-star time-dilation fraction Δ_{tot} (combining special-relativistic orbital and general-relativistic gravitational contributions, Eq. 8) as a function of compact-object mass M_X , for the 18 systems in our sample. Marker size scales with the logarithm of the orbital period. Black holes are shown as circles, neutron stars as triangles.

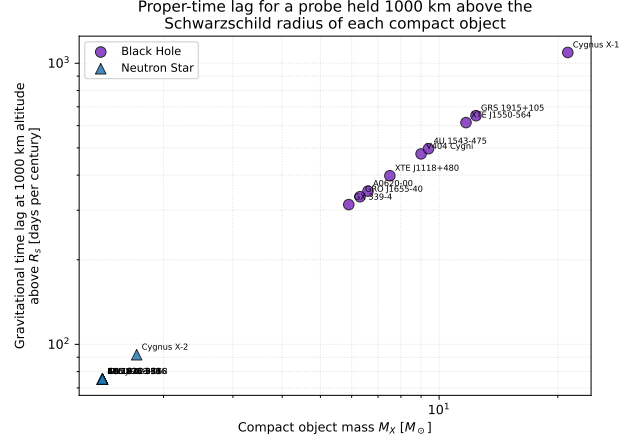


Figure 2: Gravitational time-dilation lag, expressed in days per century, for a notional probe held at a fixed altitude of 1000 km above the Schwarzschild radius R_s of each compact object, as a function of compact-object mass M_X . The lag scales approximately linearly with M_X in this regime, reaching ~ 1092 days (about 3 years) per century for Cygnus X-1.

a small but, in principle, measurable effect over sufficiently long baselines with sufficiently precise clocks.

4.3 Catalogue

4.4 Kerr black holes: spin-dependent time dilation at the ISCO

For five black holes in our sample with published continuum-fitting spin measurements — Cygnus X-1 ($a_* = 0.998^{+0.002}_{-0.0015}$, Zhao et al., 2021; Gou et al., 2011), GRS 1915+105 ($a_* = 0.98 \pm 0.02$, McClintock et al., 2006), 4U 1543-475 ($a_* = 0.80 \pm 0.05$, Shafee et al., 2006), GRO J1655-40 ($a_* = 0.70 \pm 0.10$, Shafee et al., 2006), and A0620-00 ($a_* = 0.12 \pm 0.19$, Gou et al., 2010) — we evaluate Eqs. (9) and (12) to obtain the spin-dependent ISCO radius and proper-time deficit. Figure 3 shows the resulting $1 - \tilde{E}(r_{\text{ISCO}})$ as a function of a_* , together with the Monte Carlo 68% credible intervals propagated from the published spin uncertainties.

The results span a striking range. For the near-extremal black holes, Cygnus X-1 ($a_* \approx 0.998$) and GRS 1915+105 ($a_* \approx 0.98$), the ISCO proper-time deficit reaches $\approx 32\%$ and $\approx 23\%$ respectively — roughly 4–6 times larger than the $\sqrt{8/9} \approx 5.7\%$ Schwarzschild value at the same orbital radius definition. For the moderate-spin black holes, GRO J1655-40 ($a_* \approx 0.70$) and 4U 1543-475 ($a_* \approx 0.80$), the deficit is $\approx 10\text{--}12\%$, roughly double the Schwarzschild value. For A0620-00, whose spin is consistent with zero ($a_* = 0.12 \pm 0.19$), the ISCO deficit of $\approx 6.1\%$ is close

to the Schwarzschild value, as expected.

This result demonstrates that black-hole spin is, by a wide margin, the single largest factor governing the magnitude of gravitational time dilation at the ISCO among the effects considered in this paper: spin alone can change the ISCO proper-time deficit by a factor of $\sim 5\text{--}6$, dwarfing the $\lesssim 10\%$ uncertainties from mass and distance discussed in Section 4.5. We caution that $1 - \tilde{E}(r_{\text{ISCO}})$ is not directly comparable to Δ_{1000} or Δ_{tot} from Sections 4.1–4.2, since it characterises matter in a circular orbit *at* the ISCO rather than a probe at a fixed altitude or the donor star at $a_d \gg R_s$; rather, it provides a complementary benchmark for the strong-field regime immediately outside the horizon, where accretion disks physically terminate.

4.5 Monte Carlo uncertainty propagation

Figure 4 shows the median and 68% credible interval of Δ_{tot} for all 18 systems, obtained from the Monte Carlo propagation described in Section 3.5. The relative uncertainty on Δ_{tot} (half the 16th–84th percentile range, divided by the median) ranges from $\approx 2.5\%$ for A0620-00, whose component masses are comparatively well constrained, to $\approx 40\%$ for GX 339-4, whose compact-object mass carries a large uncertainty ($M_X = 5.9^{+3.6}_{-3.6} M_\odot$; Casares and Jonker, 2014). For most systems, the relative uncertainty on Δ_{tot} is in the range 5–10%, driven primarily by the adopted 15% fractional uncertainty on donor masses without individually pub-

Table 1: Derived relativistic quantities for the 18 X-ray binary systems in our sample, ordered by compact-object type and mass. R_s is the Schwarzschild radius; Δ_{1000} is the fractional gravitational time-dilation lag for a probe at 1000 km altitude above R_s (Eq. 3); Δ_{tot} is the total donor-star time-dilation fraction (Eq. 8).

System	Type	M_X [$M_\odot$]	P_{orb} [d]	R_s [km]	Δ_{1000} [ppm]	Δ_{tot} [ppm]
Cygnus X-1	BH	21.2	5.600	62.61	29908	2.65
GRS 1915+105	BH	12.4	33.500	36.62	17823	0.39
XTE J1550-564	BH	11.7	1.542	34.55	16842	2.92
4U 1543-475	BH	9.4	1.116	27.76	13598	3.14
V404 Cygni	BH	9.0	6.473	26.58	13031	0.94
XTE J1118+480	BH	7.5	0.170	22.15	10894	9.45
A0620-00	BH	6.6	0.323	19.49	9606	5.66
GRO J1655-40	BH	6.3	2.620	18.61	9175	1.43
GX 339-4	BH	5.9	1.759	17.42	8600	1.71
Cygnus X-2	NS	1.71	9.844	5.05	2516	0.24
4U 1916-053	NS	1.40	0.0345	4.13	2061	8.94
EXO 0748-676	NS	1.40	0.158	4.13	2061	3.34
4U 1636-536	NS	1.40	0.159	4.13	2061	3.31
4U 1822-371	NS	1.40	0.233	4.13	2061	2.59
XTE J2123-058	NS	1.40	0.248	4.13	2061	2.44
Cen X-4	NS	1.40	0.629	4.13	2061	1.30
Sco X-1	NS	1.40	0.787	4.13	2061	1.14
GS 1826-238	NS	1.40	2.100	4.13	2061	0.59

lished errors and, for the neutron-star systems, the adopted $\sigma_{M_X} = 0.2 M_\odot$.

Despite these uncertainties, the ordering of systems by Δ_{tot} established in Section 4.2 is largely preserved: the credible intervals for the two most extreme systems, XTE J1118+480 and 4U 1916-053, do not overlap with those of the bulk of the sample, and the qualitative conclusion that orbital compactness (rather than compact-object mass) is the dominant driver of Δ_{tot} is robust to the propagated uncertainties. The one exception is GX 339-4, whose broad credible interval ($\Delta_{\text{tot}} \approx 0.96\text{--}2.34$ ppm at 68%) overlaps with several neighbouring systems, reflecting its poorly constrained compact-object mass.

5 Discussion

5.1 Astrophysical interpretation

The results presented here quantify, for a sample of real, dynamically characterised Galactic X-ray binaries, the magnitude of relativistic time-dilation effects accessible in the local Milky Way. Two regimes emerge clearly. First, time dilation *at a fixed altitude above the compact object* scales primarily with the compact-object’s mass through R_s (Eq. 1), making the most massive stellar black holes — exemplified by Cygnus X-1 — the most “time-dilated” locations in our sample on an absolute basis. Second, time dilation *of the donor star itself* is

governed primarily by orbital compactness (P_{orb} and hence a_d) rather than by M_X alone, since both the orbital velocity (Eq. 6) and the gravitational potential at the donor’s location (Eq. 7) increase as the orbit shrinks.

These two regimes are not in tension; rather, they reflect the fact that the donor star, no matter how compact the orbit, remains at $a_d \gg R_s$ (by 3–5 orders of magnitude in our sample), and therefore never approaches the strong-field regime in which the largest absolute time-dilation factors occur. The largest *donor-star* lags (~ 10 ppm, or several hours per century) are dramatically smaller than the largest *fixed-altitude probe* lags ($\sim 3 \times 10^{-2}$, or years per century), underscoring how steeply gravitational time dilation increases as $r \rightarrow R_s$.

5.2 Caveats and limitations

Several simplifications remain in the analysis above. First, the Schwarzschild-based calculations of Δ_{1000} and Δ_{tot} (Sections 4.1–4.2) still assume a non-rotating compact object; while Section 4.4 shows that spin can substantially enhance the proper-time deficit *at the ISCO* for the five black holes with measured a_* , we have not propagated spin into Δ_{1000} or Δ_{tot} , since these quantities are evaluated at $r \gg R_s$ where spin-induced (frame-dragging) corrections are negligible to the order considered here. Second, the donor-star calculation (Eq. 8) uses the weak-field approximation for the gravitational

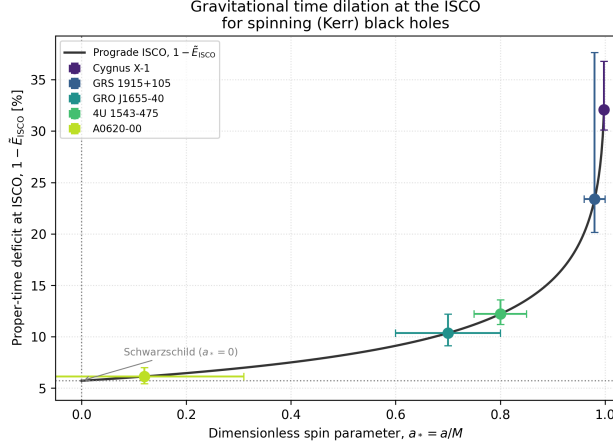


Figure 3: Fractional proper-time deficit $1 - \tilde{E}(r_{\text{ISCO}})$ for matter in a prograde circular orbit at the Kerr ISCO (Eqs. 9–12), as a function of the dimensionless spin parameter a_* . Points show the five black holes in our sample with published continuum-fitting spin measurements, with error bars representing the 68% Monte Carlo credible interval propagated from the reported spin uncertainties. The Schwarzschild value ($a_* = 0$, $1 - \tilde{E} \approx 5.7\%$) is marked for reference.

term, which is excellent given $a_d/R_s \gtrsim 10^3$ for all systems, but would break down for hypothetical ultracompact binaries not represented in our sample. Third, while Section 4.5 now propagates uncertainties in M_X and M_d through Monte Carlo sampling, we have not propagated uncertainties in P_{orb} (negligible) or i (which enters indirectly through the mass function used to derive M_X in the original dynamical studies, and is therefore already reflected in the quoted M_X uncertainties where available). Finally, all orbits are assumed circular; several systems in our sample have measured non-zero orbital eccentricities (e.g. $e \approx 0.02$ for XTE J1118+480), which would introduce a periodic modulation of Δ_{tot} over the orbital cycle, with amplitude of order e times the values reported in Table 1.

5.3 Future work

The framework presented here could be extended in several directions: (i) extending the Kerr analysis of Section 4.4 to neutron stars, for which spin periods (and hence a_*) are often more precisely known than for black holes, via millisecond pulsar timing; (ii) incorporating orbital eccentricity to compute the periodic modulation of Δ_{tot} between periastris and apoastris; (iii) extending the sample using the larger, systematically compiled catalogues of Fortin et al. (2024) and Avakyan et al. (2023), which together include well over 100 Galactic LMXBs, combined with the Monte Carlo

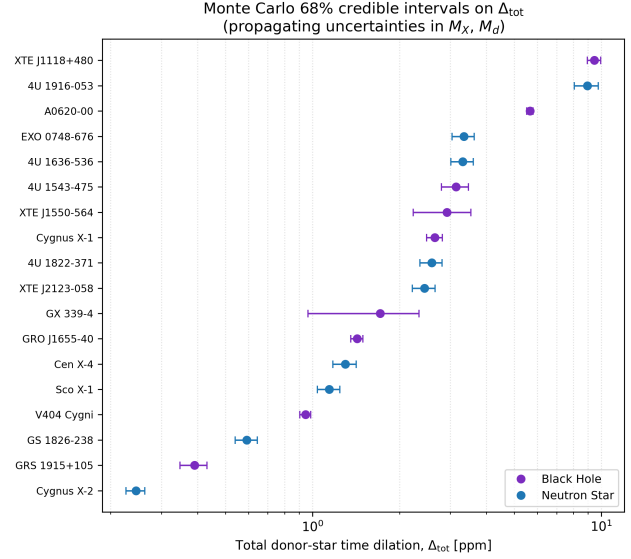


Figure 4: Median and 68% Monte Carlo credible intervals for Δ_{tot} , propagating uncertainties in M_X and M_d (see Section 3.5), for all 18 systems, sorted by median value. Black holes are shown in purple, neutron stars in blue.

framework of Section 3.5 to obtain a population-level distribution of Δ_{tot} ; and (iv) cross-matching with *Gaia* astrometric distances (Gaia Collaboration, 2023) to reduce distance-related systematics in cases where the compact-object mass depends on the assumed distance.

6 Conclusion

We have presented a catalogue of gravitational and orbital time-dilation quantities for 18 well-characterised Galactic X-ray binary systems, computed directly from published dynamical masses, orbital periods, and distances using standard general-relativistic formulae. The most massive black hole in our sample, Cygnus X-1, would impose a gravitational time lag of approximately three years per century on a probe held 1000 km above its Schwarzschild radius — by far the largest such effect accessible within the local Milky Way among the systems considered. Donor stars in the most compact binaries, such as XTE J1118+480 and 4U 1916-053, experience total relativistic time lags of up to ~ 10 ppm, equivalent to several hours per century. We further show that, for the five black holes in our sample with measured spin, rotation alone can enhance the proper-time deficit at the ISCO from the Schwarzschild value of $\sim 5.7\%$ to $\sim 32\%$ for the near-extremal Kerr black hole in Cygnus X-1 — a factor-of- ~ 6 enhancement that dwarfs the 5–10% uncertainties obtained from Monte Carlo propagation of mass and donor-mass errors for

Δ_{tot} . These results provide a concrete, observationally anchored sense of scale for relativistic time-dilation effects in the Galaxy, and the accompanying dataset and analysis code are made available for reuse and extension.

Data and Code Availability

The dataset, analysis code, and figures used in this paper are provided in the accompanying repository (`data/`, `code/`, and `figures/` directories). No proprietary data were used; all input parameters are drawn from the published literature cited in Section 2.

References

- A. Avakyan, M. Neumann, A. Zainab, V. Doroshenko, J. Wilms, et al. Xrbcats: Galactic low mass x-ray binary catalogue. *arXiv e-prints*, 2023.
- J. M. Bardeen, W. H. Press, and S. A. Teukolsky. Rotating black holes: Locally nonrotating frames, energy extraction, and scalar synchrotron radiation. *The Astrophysical Journal*, 178:347–370, 1972. doi: 10.1086/151796.
- M. E. Beer and Ph. Podsiadlowski. The quiescent light curve and the evolutionary state of gro j1655-40. *Monthly Notices of the Royal Astronomical Society*, 331:351–360, 2002. doi: 10.1046/j.1365-8711.2002.05189.x.
- J. Casares and P. G. Jonker. Mass measurements of stellar and intermediate-mass black holes. *Space Science Reviews*, 183:223–252, 2014. doi: 10.1007/s11214-013-0030-6.
- J. Casares, J. A. Orosz, C. Zurita, T. Shahbaz, et al. The spectral atlas of the neutron star x-ray binary v1727 cyg (= cygnus x-2). *The Astrophysical Journal Supplement Series*, 191:58–76, 2010. doi: 10.1088/0067-0049/191/1/58.
- J. M. Corral-Santana, J. Casares, T. Munoz-Darias, F. E. Bauer, I. G. Martinez-Pais, and D. M. Russell. Blackcat: A catalogue of stellar-mass black holes in x-ray transients. *Astronomy & Astrophysics*, 587:A61, 2016. doi: 10.1051/0004-6361/201527130.
- F. Fortin, S. Chaty, G. Tomar, J. Rodriguez, et al. A catalogue of low-mass x-ray binaries in the galaxy: From the integral to the gaia era. *Astronomy & Astrophysics*, 2024. doi: 10.1051/0004-6361/202347908.
- Gaia Collaboration. Gaia data release 3, 2023. *Astronomy & Astrophysics*, 674, A1.
- D. K. Galloway, M. P. Munro, J. M. Hartman, D. Psaltis, and D. Chakrabarty. Thermonuclear (type i) x-ray bursts observed by the rossi x-ray timing explorer. *The Astrophysical Journal Supplement Series*, 179:360–422, 2008. doi: 10.1086/592044.
- L. Gou, J. E. McClintock, J. Liu, R. Narayan, et al. A determination of the spin of the black hole primary in lmc x-3. *The Astrophysical Journal*, 718:620–634, 2010. doi: 10.1088/0004-637X/718/2/620.
- L. Gou, J. E. McClintock, M. J. Reid, J. A. Orosz, et al. The extreme spin of the black hole in cygnus x-1. *The Astrophysical Journal*, 742:85, 2011. doi: 10.1088/0004-637X/742/2/85.
- Q. Z. Liu, J. van Paradijs, and E. P. J. van den Heuvel. A catalogue of low-mass x-ray binaries in the galaxy, lmc, and smc (fourth edition). *Astronomy & Astrophysics*, 469:807–810, 2007. doi: 10.1051/0004-6361:20077303.
- J. E. McClintock, R. Shafee, R. Narayan, R. A. Remillard, S. W. Davis, and L.-X. Li. The spin of the near-extreme kerr black hole grs 1915+105. *The Astrophysical Journal*, 652:518–539, 2006. doi: 10.1086/508457.
- J. C. A. Miller-Jones, P. G. Jonker, V. Dhawan, W. Briskin, et al. The first accurate parallax distance to a black hole. *The Astrophysical Journal Letters*, 706:L230–L234, 2009. doi: 10.1088/0004-637X/706/2/L230.
- J. C. A. Miller-Jones, A. Bahramian, J. A. Orosz, I. Mandel, et al. Cygnus x-1 contains a 21-solar mass black hole—implications for massive star winds. *Science*, 371:1046–1049, 2021. doi: 10.1126/science.abb3363.
- C. W. Misner, K. S. Thorne, and J. A. Wheeler. *Gravitation*. W. H. Freeman, 1973.
- M. J. Reid, J. E. McClintock, J. F. Steiner, D. Steeghs, et al. A parallax distance to the microquasar grs 1915+105 and a revised estimate of its black hole mass. *The Astrophysical Journal*, 796:2, 2014. doi: 10.1088/0004-637X/796/1/2.
- R. Shafee, J. E. McClintock, R. Narayan, S. W. Davis, L.-X. Li, and R. A. Remillard. Estimating the spin of stellar-mass black holes by spectral fitting of the x-ray continuum. *The Astrophysical Journal Letters*, 636:L113–L116, 2006. doi: 10.1086/498938.
- S. L. Shapiro and S. A. Teukolsky. *Black Holes, White Dwarfs, and Neutron Stars: The Physics of Compact Objects*. Wiley, 1983.

- D. Steeghs, J. E. McClintock, S. G. Parsons, M. J. Reid, S. Littlefair, and V. S. Dhillon. The not-so-massive black hole in the microquasar grs 1915+105. *The Astrophysical Journal*, 768:185, 2013. doi: 10.1088/0004-637X/768/2/185.
- X. Zhao, L. Gou, Y. Dong, Y. Tuo, et al. Re-estimating the spin parameter of the black hole in cygnus x-1. *The Astrophysical Journal*, 916:108, 2021. doi: 10.3847/1538-4357/abff5a.